

Course: Linear Algebra

Subject Teacher: Gunesh Kumar



Department of Computer Science,
Sindh Madressatul Islam University, Karachi.

Vector Space or Linear Space

Basic Introduction: Vector Space is the generalization of ordinary vectors or physical space. For Example: A vector of the form $\vec{u} = [u_1, u_2, u_3]$ or $\vec{v} = [v_1, v_2, v_3]$. In simple words we say that these two vectors $\vec{u}$ and $\vec{v}$ form the vector space in R^3 .

In the general; A vector of the form $\vec{u} = [u_1, u_2, u_3, \dots u_n]$ is called vector space in R^n and $u_1, u_2, u_3, \dots u_n$ are the elements or components of the vector.

Vector Space or Linear Space

Of course we can not see or sketch vectors in R^5 , R^6 or R^n as we can in R^2 or R^3 . But we can imagine various mathematical or physical situations in which they are useful.

General concept of vector space: The generalized set of vector space can be (sets of vectors, functions, matrices, operators, polynomials etc.) that play very important role in Applied Mathematics.

Definition of Vector Space

Let $\vec{V}$ be a set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots \dots \dots v_n\}$ then following axioms are satisfied for vector space:

Axioms of Additions

- (i). Closure law over vector addition
$$\vec{V}_1 + \vec{V}_2 \in \vec{V}$$
- (ii). Commutative law over vector addition
$$\vec{V}_1 + \vec{V}_2 = \vec{V}_2 + \vec{V}_1$$
- (iii). Associative law over vector addition
$$(\vec{V}_1 + \vec{V}_2) + \vec{V}_3 = \vec{V}_1 + (\vec{V}_2 + \vec{V}_3)$$
- (iv). Existence of additive identity
$$\vec{V}_1 + \vec{O} = \vec{V}_1 \quad \text{or} \quad \vec{O} + \vec{V}_1 = \vec{V}_1$$
- (v). Existence of additive inverse
$$\vec{V}_1 + (-\vec{V}_1) = \vec{O} \quad \text{or} \quad (-\vec{V}_1) + \vec{V}_1 = \vec{O}$$

Definition of Vector Space

Axioms of Multiplication

- (i). Closure law over scalar multiplication
$$K\overrightarrow{V_1} \in \overrightarrow{V}$$
- (ii). Distributive law over scalar multiplication
$$K(\overrightarrow{V_1} + \overrightarrow{V_2}) = K\overrightarrow{V_2} + K\overrightarrow{V_1}$$
- (iii). Distributive law over scalar addition
$$(K + L)\overrightarrow{V_1} = K\overrightarrow{V_1} + L\overrightarrow{V_1}$$
- (iv). Associative law over scalar multiplication
$$K(L\overrightarrow{V_1}) = L(K\overrightarrow{V_1})$$
- (v). Existence of multiplicative identity
$$1.\overrightarrow{V_1} = \overrightarrow{V_1} \quad \text{or} \quad \overrightarrow{V_1}.1 = \overrightarrow{V_1}$$

Some Important Points

Vector space V (Set of vector) is also called linear space.

Vectors V (Set of vector) may be any mathematical objects, for example: real number, complex numbers, matrices, functions, polynomial, etc.

Vector space can be written as (V, F) or (F) , where F is any mathematical object.

From axioms of additions (i) to (v), It is obvious that V (Set of vector) is Abelian group under vector addition.

Topics in Vector Space

Linear combination of Vectors

Linear dependence and independence of vectors

Spanning of vectors

Basis in vector space

Dimension of vector space

Nullity in vector space

Linear Combination

Let w be a vector space having the set of vectors $v_1, v_2, v_3, \dots, v_n$ is to be linear combination if it satisfies the condition below

$$w = c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_n v_n$$

Here our aim is to find the values of constants

$$c_1, c_2, c_3, \dots c_n.$$

Example:1. Write the vector $w = (1, -2, 5)$ as linear combination of vectors in the set of vectors $S = (s_1, s_2, s_3)$ given by

$$S = \{(1, 1, 1), (1, 2, 3), (2, -1, 1)\}.$$

Solution

The linear combination for $w = (1, -2, 5)$ in terms of v_1, v_2, v_3 can be obtained from vector equation

$$\mathbf{w} = \mathbf{c}_1 \mathbf{v}_1 + \mathbf{c}_2 \mathbf{v}_2 + \mathbf{c}_3 \mathbf{v}_3 \quad (1)$$

$$(1, -2, 5) = c_1(1, 1, 1) + c_2(1, 2, 3) + c_3(2, -1, 1)$$

$$(1, -2, 5) = (c_1, c_1, c_1) + (c_2, 2c_2, 3c_2) + (2c_3, -c_3, c_3)$$

Writing components combinely,

$$(1, -2, 5) = (c_1 + c_2 + 2c_3), (c_1 + 2c_2 - c_3), (c_1 + 3c_2 + c_3)$$

By equating corresponding components, we get the system of linear equations

$$c_1 + c_2 + 2c_3 = 1 \quad (i)$$

$$c_1 + 2c_2 - c_3 = -2 \quad (ii)$$

$$c_1 + 3c_2 + c_3 = 5 \quad (iii)$$

Solving above system of linear equations to find c_1 , c_2 and c_3 .

So converting system of linear equations into matrix form

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & -1 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

Before solving matrix equations using Gauss-Elimination method, we construct augmented matrix as

$$A_b = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & -1 & -2 \\ 1 & 3 & 1 & 5 \end{bmatrix}$$

Now, compute Echelon form by using row elementary operations

$$A_b = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{bmatrix}, \quad R_2 - R_1, R_3 - R_1, R_3 - 2R_2$$

Writing in terms of linear equations

$$c_1 + c_2 + 2c_3 = 1 \quad (i)$$

$$c_2 - 3c_3 = -3 \quad (ii)$$

$$5c_3 = 10 \quad (iii)$$

Solving above equations, we get

$$c_1 = -6, \quad c_2 = 3 \quad \text{and} \quad c_3 = 2$$

As we know that, $\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$

Putting values of constants which we found viz c_1, c_2, c_3 in above equation. We get

$$(1, -2, 5) = -6\mathbf{v}_1 + 3\mathbf{v}_2 + 2\mathbf{v}_3$$

is the required Linear Combination of vectors.

Verification of solutions

Since, $w = c_1 v_1 + c_2 v_2 + c_3 v_3$

Putting values of w, c_1, c_2, c_3 and v_1, v_2, v_3 in above equation

$$(1, -2, 5) = -6(1, 1, 1) + 3(1, 2, 3) + 2(2, -1, 1)$$

$$\text{We get, } (1, -2, 5) = (1, -2, 5)$$

Hence, Linear Combination is correct.

Note:

(i). Set of vectors can be expressed as a Linear Combination if system of Linear equations is consistent viz $|A| \neq 0$.

(ii). Set of vectors can not be expressed as a Linear Combination if system of Linear equations is inconsistent viz $|A| = 0$.

Example:2. Write the vector $w = (1, -2, -2)$ as linear combination of vectors (if possible) in the set of vectors $S = (s_1, s_2, s_3)$ given by
$$S = \{(1, 2, 3), (0, 1, 2), (-1, 0, 1)\}.$$

Solution

The linear combination for $w = (1, -2, -2)$ in terms of v_1, v_2, v_3 can be obtained from vector equation

$$\mathbf{w} = \mathbf{c}_1 \mathbf{v}_1 + \mathbf{c}_2 \mathbf{v}_2 + \mathbf{c}_3 \mathbf{v}_3 \quad (1)$$

$$(1, -2, -2) = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-1, 0, 1)$$

$$c_1 - c_3 = 1 \quad (i)$$

$$2c_1 + c_2 = -2 \quad (ii)$$

$$3c_1 + 2c_2 + c_3 = 2 \quad (iii)$$

Solving above system of linear equations to find c_1, c_2 and c_3 .

So converting system of linear equations into matrix form

$$AX = B$$
$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$$

Set of vectors can not be expressed as a Linear Combination if system of Linear equations is inconsistent viz $|A| = 0$.

Assignment Work

Example:1. In R^3 express the vector $w = (2, -5, 3)$ as linear combination of vectors (if possible) in the set of vectors $V = (v_1, v_2, v_3)$ given by

$$V = \{(1, -3, 2), (2, -4, -1), (1, -5, 7)\}.$$

Example:2. Given the vector $w = (-1, -2, -2)$, express it as linear combination of vectors $v_1 = (0, 1, 4)$, $v_2 = (-1, 1, 2)$ and $v_3 = (3, 1, 2)$.

Example:3. Express the vector $w = (1, 1, 1)$ as linear combination of vectors $p_1 = (1, 2, 3)$, $p_2 = (0, 1, 2)$ and $p_3 = (-1, 0, 1)$.

Linear Dependence and Independence of Vectors

Definition: Let the set of vectors $v_1, v_2, v_3, \dots, v_n$ then vectors can be checked for linearly dependence and independence by the below condition:

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + \dots + c_nv_n = 0$$

Procedure

For linearly dependent and independent vectors

1. Form the vector's equation

$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + \dots + c_nv_n = 0$ and write a homogenous system of linear equations in terms of $c_1, c_2, c_3, \dots, c_n$.

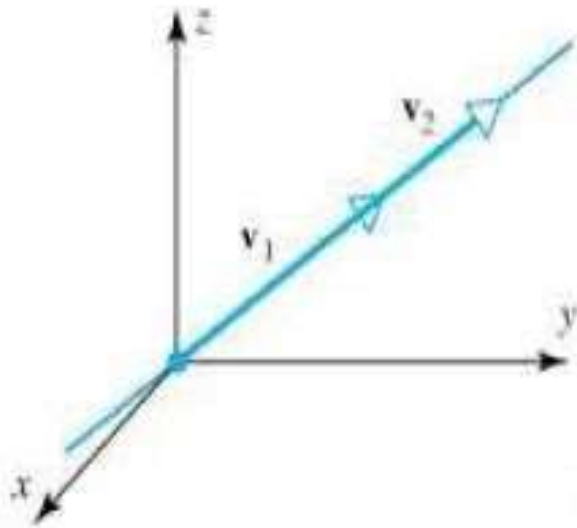
2. By applying the Gaussian – Elimination method to find constant then two conditions will arise:

(a). If the system of linear equations has non-trivial solution means $(c_1 \neq 0, c_2 \neq 0, c_3 \neq 0, \dots, c_n \neq 0)$ then set of vectors is linearly dependent.

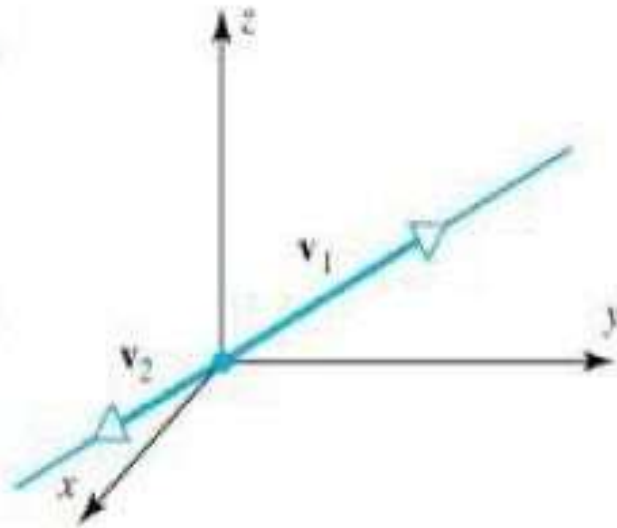
(a). If the system of linear equations has trivial solution means $(c_1 = 0, c_2 = 0, c_3 = 0, \dots, c_n = 0)$ then set of vectors is linearly independent

Geometry of Linearly Dependent and Independent Vectors

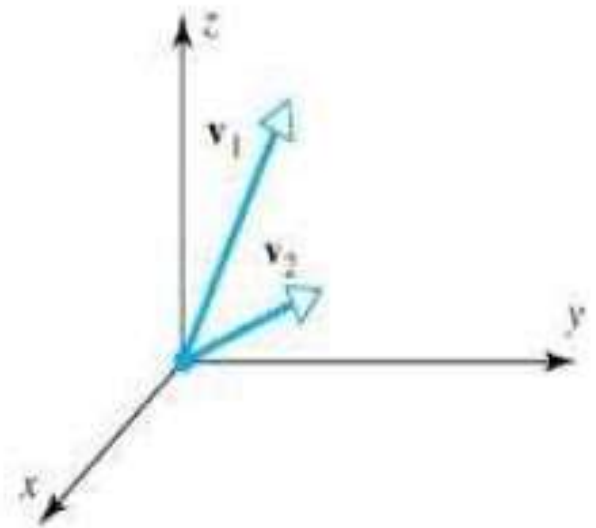
A set of vectors is said to be linearly dependent if and only if the vectors lie on the same line; otherwise, they are linearly independent.



(a) Linearly dependent



(b) Linearly dependent



(c) Linearly independent

Example:1. Determine whether the set of vectors in R^3 is linearly dependent or linearly independent given by $S = \{(1,2,3), (0,1,2), (-2,0,1)\}$.

Solution: Form the vector equation

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad (1)$$

$$c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (0,0,0)$$

$$(c_1 + 0c_2 - 2c_3, 2c_1 + c_2 + 0c_3, 3c_1 + 2c_2 + c_3) = (0,0,0)$$

By equating each component, we get following homogenous system of linear equations

$$c_1 - 2c_3 = 0 \quad (i)$$

$$2c_1 + c_2 = 0 \quad (ii)$$

$$3c_1 + 2c_2 + c_3 = 0 \quad (iii)$$

Writing system of linear equations into matrix form

$$AX = 0$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Reduce A matrix into Echelon or Reduced echelon form using elementary row operations,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ The operations are}$$

$$R_2 - 2R_1, R_3 - 3R_1, R_3 - 2R_2, (-1) R_3, R_2 - 4R_3, R_1 + 2R_3.$$

This show that $c_1=c_2=c_3=0$, Since the system has trivial solution, Hence the given vectors are linearly independent.

Note:

- (i). Set of vectors are said to be linearly dependent if $|A| = 0$.
- (ii). Set of vectors are said to be linearly independent if $|A| \neq 0$.

Example:2. Determine whether the set of vectors in R^3 is linearly dependent or linearly independent given by $S = \{(1,0,0), (0,1,0), (0,0,1)\}$.

Solution: Form the vector equation

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad (1)$$

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (0,0,0)$$

$$(c_1 + 0c_2 + 0c_3, 0c_1 + c_2 + 0c_3, 0c_1 + 0c_2 + c_3) = (0,0,0)$$

By equating each component, we get following homogenous system of linear equations

$$c_1 = 0, c_2 = 0, c_3 = 0.$$

Since the system has trivial solution, vectors are linearly independent.

Example:3. Prove that the set of vectors
 $S = \{(1, -1, 2), (2, 3, -4), (0, 5, -8)\}$ is linearly dependent

Solution: Form the vector equation

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad (1)$$

$$c_1(1, -1, 2) + c_2(2, 3, -4) + c_3(0, 5, -8) = (0, 0, 0)$$

$$(c_1 + 2c_2 + 0c_3, -c_1 + 3c_2 + 5c_3, 2c_1 - 4c_2 - 8c_3) = 0$$

By equating each component, we get following homogenous system of linear equations

$$c_1 + 2c_2 = 0 \quad (i)$$

$$-c_1 + 3c_2 + 5c_3 = 0 \quad (ii)$$

$$2c_1 - 4c_2 - 8c_3 = 0 \quad (iii)$$

Writing system of linear equations into matrix form

$$AX = 0$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 5 \\ 2 & -4 & -8 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Reduce A matrix into Echelon or Reduced echelon form using elementary row operations,

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \text{ The operations are}$$

$$R_2 + R_1, R_3 - 2R_1, \frac{1}{5}R_2, -\frac{1}{8}R_3.$$

By equating each component, we get following homogenous system of linear equations

$$c_1 + 2c_2 = 0 \quad (iv)$$

$$c_2 + c_3 = 0 \quad (v)$$

$$c_2 + c_3 = 0 \quad (vi)$$

Put $c_3 = k$ in equation (vi), we get $c_2 = -k$. Put $c_2 = -k$ in equation (v), we get $c_3 = k$, in this way, we get

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2k \\ -k \\ k \end{bmatrix}$$

It has infinite solutions for $k = 0, 1, 2, 3, 4, 5, \dots$. Means, we got non-trivial solution, hence vectors are linearly dependent.

Assignment Work

Check whether the following set of vectors for linearly dependence and independence given as:

(i). $v_1 = (2, 4, -3), v_2 = (0, 1, 1), v_3 = (0, 1, -1).$

(ii). $u_1 = (1, 2, 3), u_2 = (0, 1, 2), u_3 = (-2, 0, 1).$

22. Here are three vectors. Determine whether they are linearly independent or linearly dependent.

$$\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$$

23. Here are three vectors. Determine whether they are linearly independent or linearly dependent.

$$\begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

24. Here are three vectors. Determine whether they are linearly independent or linearly dependent.

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$$

25. Here are four vectors. Determine whether they span $\mathbb{R}^3$. Are these vectors linearly independent?

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

Spanning of Vector

If $\vec{u} = \{v_1, v_2, v_3, \dots \dots \dots v_n\}$ is a set of vectors in a vector space V , then **the span of v** is the set of all linear combinations of the vectors in v is called spanning of a set. Following formula can be used to span the set:

$$span(u) = \{c_1v_1 + c_2v_2 + c_3v_3 + \square + c_kv_k\}$$

Simply it tells us whether the set of vectors can be expressed as a linear combination or not.

Procedure

Step (1). Let an arbitrary vector $u = (u_1, u_2, u_3)$.

Step (2). Apply the condition for spanning

$$u = (c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_n v_n)$$

Step (3). Decision

If $\det(A) \neq 0$ then system has solution and vectors can be expressed for linear combination.

If $\det(A) = 0$ then system has not solution and vectors can not be expressed for linear combination.

Example : 01. Show that the set of vectors
 $V = \{(1,2,3), (0,1,2), (-2,0,1)\}$ spans in R^3 .

Solution:

Let the set of given vectors

$$v_1 = (1,2,3), v_2 = (0,1,2) \text{ and } v_3 = (-2,0,1).$$

Step (1). Let an arbitrary vector $u = (u_1, u_2, u_3)$.

Step (2). Apply the condition for spanning

$$u = (c_1 v_1 + c_2 v_2 + c_3 v_3 + \cdots + c_n v_n)$$

Or

$$u_1, u_2, u_3 = c_1 v_1 + c_2 v_2 + c_3 v_3 \quad (1)$$

Putting values in equation (1)

$$u_1, u_2, u_3 = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-2, 0, 1)$$

By equating, we get

$$c_1 + 0c_2 - 2c_3 = u_1$$

$$2c_1 + c_2 + 0c_3 = u_2$$

$$3c_1 + 2c_2 + c_3 = u_3$$

Converting into matrix form, $AX = B$

$$\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

Since $|A| \neq 0$. So, Set of vectors can be expressed for linear combination. Hence, set of vectors is spanning in R^3 .

Example : 01. Show that the set of vectors
 $V = \{(1,1,2), (1,0,1), (2,1,3)\}$ spans in R^3 .

Solution:

Let the set of given vectors

$$v_1 = (1,1,2), v_2 = (1,0,1) \text{ and } v_3 = (2,1,3).$$

Step (1). Let an arbitrary vector $u = (u_1, u_2, u_3)$.

Step (2). Apply the condition for spanning

$$u = (c_1 v_1 + c_2 v_2 + c_3 v_3 + \cdots + c_n v_n)$$

Or

$$u_1, u_2, u_3 = c_1 v_1 + c_2 v_2 + c_3 v_3 \quad (1)$$

Putting values in equation (1)

$$u_1, u_2, u_3 = c_1(1,1,2) + c_2(1,0,1) + c_3(2,1,3)$$

By equating, we get

$$c_1 + c_2 + 2c_3 = u_1$$

$$c_1 + 0c_2 + c_3 = u_2$$

$$2c_1 + c_2 + 3c_3 = u_3$$

Converting into matrix form, $AX = B$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

Since $|A| = 0$. So, Set of vectors can not be expressed for linear combination. Hence, set of vectors is not spanning in R^3 .

Assignment Work

Example : 01. Show that the set of vectors
 $V = \{ (0,1,2), (0,2,3), (0,3,1) \}$ spans in R^3 .

Thanks